Measuring the Frequency of Errors

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Abstract. We study the problem of *distribution-aware reliability estimation*: given a deterministic program P, a discrete distribution $\mathcal{D}$ over its inputs, and a Boolean property φ, certify a confidence interval for $\Pr_{x\sim\mathcal{D}}[\varphi(P(x)) = 1]$ using as few program evaluations as possible. The central difficulty is that rare-event programs defeat plain sampling, while symbolic techniques fail on programs with non-linear arithmetic or unbounded loops.

We present DiSE, an adaptive partitioning algorithm that bridges the two regimes. DiSE maintains a frontier of path-condition regions and at each step chooses between sampling and SMT-driven refinement under a unified gain-per-cost criterion. The certified half-width decomposes as $\Pr[|\hat{\mu} - \mu| \leq \varepsilon_{\text{stat}} + W_{\text{open}}] \geq 1 - \delta$, where $\varepsilon_{\text{stat}}$ aggregates per-region statistical uncertainty and W_{open} is the total probability mass of regions not yet symbolically certified. This decomposition is both the organising principle of the algorithm and the content of its central correctness theorem: every action reduces one term, and the algorithm terminates when their sum falls below ε.

The per-region statistical bound uses the predictable-plug-in empirical-Bernstein confidence sequence of Waudby-Smith and Ramdas — closed-form, variance-adaptive, and anytime-valid under DiSE's data-dependent stopping. The closure rule is single-branch: a region is certified if and only if an SMT oracle proves path-determinism, yielding a one-lemma soundness contract with no sample-based fallback.

We evaluate DiSE, a Python prototype with a Z3 backend, and evaluate it on 12 integer benchmarks. On rare-event programs whose path structure aligns with the input distribution, DiSE reduces sample counts by up to two orders of magnitude compared with plain and stratified Monte Carlo.

Keywords: Probabilistic program analysis · Statistical model checking · Reliability estimation · Symbolic execution · Anytime-valid confidence intervals.

1 Introduction

Consider a network packet-routing function deployed on a server handling millions of requests per day. The function computes a hash of each packet's header and maps it to one of k queues; a collision sends two packets to the same queue, degrading throughput. The production traffic distribution is far from uniform—certain header patterns are orders of magnitude more common than others. The

engineering team knows that edge-case collisions exist, because a fuzzer found one. What they need to know is the *rate*: what fraction of production inputs actually trigger the defect? A single counterexample does not answer this question.

This scenario is an instance of *distribution-aware reliability estimation*. Given a deterministic program $P : \mathcal{X} \to \mathcal{Y}$, a discrete distribution $\mathcal{D}$ over $\mathcal{X}$, and a Boolean property $\varphi : \mathcal{Y} \to \{0, 1\}$, the *operational reliability* of P under $\mathcal{D}$ with respect to φ is

$$\mu \;=\; \Pr_{x \sim \mathcal{D}}\big[\varphi(P(x)) = 1\big]. \tag{1}$$

Reliability in this sense arises in *operational testing*, where a developer must certify that a production input distribution produces a failure rate below a contractual threshold; in *probabilistic program verification* [?,?], where the question is inherently quantitative; and in *distribution-aware property-based testing* [?,?], where quantitative coverage supplements counterexample search.

The challenge. Three difficulties set reliability estimation apart from conventional program analysis.

The answer is a real number, not a Boolean. Standard verification asks $\forall x\colon \varphi(P(x))$. Reliability asks for the *measure* of the failure set $\{x : \varphi(P(x)) = 1\}$ under $\mathcal{D}$. Counterexample-driven techniques produce a single witness; bounding the probability of the entire set requires a different methodology.

The rare-event regime matters most and is hardest. When $\mu \ll 1$, plain Monte Carlo requires $\Theta(1/(\mu\varepsilon^2))$ samples to certify a half-width of ε—a prohibitive budget when $\mu = 10^{-4}$ and $\varepsilon = 10^{-5}$. Yet rare events are precisely where operational reliability matters: a defect that occurs once per million inputs under a uniform distribution may occur once per hundred inputs under production traffic.

The certificate is the deliverable. A point estimate for μ without an accompanying confidence interval is operationally useless: one cannot determine whether $\hat{\mu} = 0.003$ is consistent with a contractual bound of ≤ 0.001 unless the estimator's uncertainty is quantified. Sound, tight intervals—not just point estimates—are the engineering artifact that reliability estimation must produce.

Existing approaches. Two families of techniques address this problem, each with complementary strengths and limitations.

Probabilistic symbolic execution (PSE) [?,?,?] enumerates program paths, encodes each path condition as a formula, and computes its exact probability via model counting [?,?]. PSE is exact where it applies. However, path explosion forces incompleteness on programs with unbounded loops or non-linear arithmetic, and non-uniform input distributions require ad-hoc extensions of the model-counting backend that are not always available.

Plain sampling [?,?] draws iid inputs, executes the program, and applies a concentration inequality to the empirical failure rate. Sampling is robust to all sources of PSE difficulty, but inherits the $\Theta(1/\varepsilon^2)$ rate of independent estimation. In the rare-event regime, the sample budget required to reach a useful half-width is simply too large.

Our approach. We propose a middle course between exact symbolic enumeration and unguided sampling. DiSE maintains a *frontier*: a tree whose leaves partition the input domain $\mathcal{X}$ into path-condition regions. At each step the algorithm chooses between two actions: $\textsc{Sample}(\pi)$, which draws additional executions from a region π to tighten its statistical confidence bound, and $\textsc{Refine}(\pi, b)$, which invokes an SMT solver to split π along a branch predicate b harvested from observed executions, potentially certifying that an entire child region is all-pass or all-fail under φ. The choice is governed by a single gain-per-cost criterion applied uniformly to both action types.

The key insight is a two-part decomposition of the certified half-width:

$$\Pr\big[\,|\hat{\mu} - \mu| \;\le\; \varepsilon_{\text{stat}} + W_{\text{open}}\,\big] \;\ge\; 1 - \delta, \tag{2}$$

where $\varepsilon_{\text{stat}}$ aggregates per-region statistical uncertainty and W_{open} is the total probability mass of regions not yet symbolically certified. Every $\textsc{Sample}$ action reduces $\varepsilon_{\text{stat}}$; every successful $\textsc{Refine}$ reduces W_{open}. The algorithm terminates as soon as $\varepsilon_{\text{stat}} + W_{\text{open}} \le \varepsilon$. There are no phases, no hand-tuned switching heuristics: the decomposition (2) is simultaneously the correctness argument and the scheduler.

Two design choices distinguish DiSE from prior work. First, the per-region statistical bound instantiates the predictable-plug-in empirical-Bernstein (PrPl-EB) confidence sequence of Waudby-Smith and Ramdas [?], which is closed-form, variance-adaptive, and *anytime-valid*: it remains sound under DiSE's data-dependent stopping rule, strictly tightening the union-bounded Wilson construction used in earlier statistical model checking tools [?,?]. Second, the closure rule is single-branch: a region is certified if and only if the SMT backend proves path-determinism ($F_\pi \wedge \neg\,\text{path}_\pi$ is unsat). There is no sample-based fallback. The price is wider intervals on programs whose closure is symbolically intractable; the gain is a one-lemma soundness contract and the elimination of a class of silent bias that affects prior prototypes.

Contributions. This paper makes three contributions.

1. *The* DiSE *algorithm and its correctness proof.* We introduce DiSE and prove that its reported interval $[L, U]$ covers μ with probability at least $1 - \delta$ (Section 1). The proof reduces to two independently checkable invariants—mass-conservative refinement (Section 1) and SMT-certified closure (Section 2)—plus an application of an anytime-valid concentration inequality per leaf.
2. *Anytime-valid certification via PrPl-EB.* We give the first application of the predictable-plug-in empirical-Bernstein confidence sequence in the context of program reliability estimation. The bound is closed-form, variance-adaptive, and requires no prior knowledge of per-leaf variance, strictly tightening standard Hoeffding and Wilson constructions under adaptive stopping.
3. *Implementation and evaluation.* We implement DiSE as DiSE, a Python prototype with a Z3 backend and a benchmark suite of 12 integer programs. On rare-event benchmarks whose path structure aligns with the input distribution, DiSE reduces sample counts by up to two orders of magnitude relative to plain and stratified Monte Carlo.

Paper organisation. Section 2 formally states the problem and defines the input distribution class. Section 3 surveys related work. Section 4 presents DiSE and proves Section 1. Section 5 describes the DiSE implementation. Section 6 reports experimental results. Section 7 concludes.

2 Problem Statement

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a finite input domain $\mathcal{X} \subseteq \mathbb{Z}^d$. Throughout the paper we make the following assumption on the program.

Assumption. *The program $P : \mathcal{X} \to \mathcal{Y}$ is* deterministic*: the output $P(x)$ is determined entirely by the input x, with no dependence on any internal source of randomness. Programs that use randomness must be wrapped to expose the random seed as an explicit input.*

We consider input distributions drawn from the following class.

Definition 1 (Product-form discrete distributions, class (D1)). *A distribution $\mathcal{D}$ over $\mathcal{X} = \mathbb{Z}^d$ is in class* (D1) *if it factors as $\mathcal{D}(x_1, \ldots, x_d) = \prod_{i=1}^{d} \mathcal{D}_i(x_i)$, where each marginal $\mathcal{D}_i$ has a closed-form cumulative probability-mass function over integer intervals: $\mathcal{D}_i(\{a, \ldots, b\}) = \mathcal{D}_i(b) - \mathcal{D}_i(a-1)$. Uniform, bounded geometric, categorical, truncated Poisson, and finite Bernoulli mixtures all belong to (D1).*

The (D1) restriction is the source of DiSE's central variance-reduction mechanism. For any axis-aligned box $R = \prod_i [a_i, b_i]$, the probability $\Pr_{\mathcal{D}}[X \in R] = \prod_i \mathcal{D}_i(\{a_i, \ldots, b_i\})$ is available in closed form and carries *zero estimation variance*. Bayes-net-structured and general discrete distributions fall outside this class and are left to future work (Section 7).

Let $\varphi : \mathcal{Y} \to \{0, 1\}$ be a Boolean output property. The operational reliability $\mu \in [0, 1]$ is defined as in (1).

Problem definition. Given P, $\mathcal{D} \in$ (D1), φ, a target half-width $\varepsilon \in (0, 1]$, and a confidence parameter $\delta \in (0, 1)$, produce an estimator $\hat{\mu}$ and a two-sided interval $[L, U] \subseteq [0, 1]$ satisfying the *soundness contract*

$$\mathbb{P}\big[\, L \le \mu \le U \,\big] \;\ge\; 1 - \delta, \qquad U - L \;\le\; 2\varepsilon, \tag{3}$$

using as few evaluations of P as possible. The probability in (3) is over the algorithm's internal randomness and the iid samples drawn from $\mathcal{D}$; the true reliability μ is a fixed constant.

Assertion-violation framing. The output-property formulation (1) subsumes the classical assertion-violation framing. For a program annotated with an `assert` statement, the failure probability $\Pr_{\mathcal{D}}[P \text{ raises } \mathsf{AssertionError}]$ is recovered by wrapping P in a handler that converts the targeted exception into a Boolean failure flag and setting $\varphi(y) = [y = 1]$. Worked examples of both framings appear in Section 6.

Why the problem is hard. DiSE addresses three inherent difficulties, each of which is mitigated by a distinct component of its design.

1. *Real-valued answer.* Reliability is a measure, not a predicate. DiSE maintains a fractional estimator over a partition of $\mathcal{X}$, accumulating evidence from both sampling and symbolic reasoning.
2. *Rare events.* When $\mu \approx 0$ or $\mu \approx 1$, plain Monte Carlo is sample-inefficient. DiSE stratifies the input domain by path condition, concentrating samples in regions of high variance and using closed-form mass computations in certified regions.
3. *Sound, tight intervals.* A point estimate without a certificate is operationally useless. DiSE carries an explicit anytime-valid confidence bound at every step, so the interval it reports is sound regardless of when it stops.

3 Related Work

DiSE sits at the intersection of five research lines. We position it against each.

Probabilistic symbolic execution. Geldenhuys, Dwyer, and Visser [?] introduced probabilistic symbolic execution (PSE), combining symbolic execution with model counting (LattE, Barvinok [?]) to compute exact path probabilities for affine path conditions over uniform inputs. Filieri et al. [?] extended PSE to structured data; Borges et al. [?] introduced compositional model counting. Subsequent work, surveyed in Pasareanu's synthesis lectures [?], has broadened the class of treatable path conditions and input distributions.

PSE is exact where it applies; DiSE is statistical and produces $(1-\delta)$-coverage intervals. On programs where PSE applies—bounded linear-integer-arithmetic paths and tractable input distributions—PSE strictly dominates. DiSE fills the gap when programs have unbounded loops, non-linear arithmetic, or non-uniform input distributions. When every region in DiSE's frontier admits a closed-form mass and the closure rule succeeds on every leaf, the two approaches agree exactly.

Statistical and probabilistic model checking. PRISM [?] and Storm [?] verify quantitative temporal properties on Markov chains and MDPs. Statistical variants—PLASMA-Lab [?], the Modest Toolset [?], and Storm's SMC front-end [?]—discharge the same specifications by sampling the underlying model. These tools operate on dedicated modelling languages (PRISM, JANI, Modest); DiSE operates directly on Python source, treating the program itself as the model. The trade-off is complementary: DiSE cannot reason about nondeterminism or scheduler choices, while PMC tools cannot reason about source-level loops without a hand-built model.

Probabilistic assertions. Sampson et al. [?] introduced *passerts*—probabilistic assertions in C programs verified by sampling with Hoeffding or Bayesian bounds. DiSE's `failure_probability` API targets the same query class. The difference

is the bound and the sampling strategy: passerts use plain Hoeffding sampling over the whole input space, whereas DiSE stratifies by symbolic path condition and uses a variance-adaptive anytime-valid bound [?]. On programs whose path structure concentrates probability mass, DiSE reaches the same certified half-width with substantially fewer samples; on programs with flat, well-mixed distributions the advantage narrows.

Model counting. Approximate model counting [?,?] provides PAC-(ε, δ) counts for propositional satisfiability problems. Recent work has added formal certificates in Isabelle [?]. DiSE does not currently invoke a model counter—it uses importance sampling to estimate mass on non-axis-aligned regions—but a natural extension would delegate the region-mass subproblem to a certified counter, retaining $(1 - \delta)$-coverage with tighter constants.

Anytime-valid concentration inequalities. Classical Wilson and Hoeffding bounds are valid only at a *fixed* sample size; applying them under data-dependent stopping requires a union bound over sample counts. The mixture-martingale literature, culminating in Howard et al. [?], provides time-uniform Chernoff bounds via non-negative supermartingales. Waudby-Smith and Ramdas [?] sharpen this with a closed-form predictable-plug-in empirical-Bernstein (PrPl-EB) confidence sequence that is variance-adaptive and matches the asymptotic width of Bennett's inequality without requiring the variance to be known. DiSE uses PrPl-EB per leaf with Bonferroni correction over the (bounded) leaf count.

Property-based testing. QuickCheck [?] and Hypothesis [?] search for a single input that violates a property—a different goal from bounding the failure rate. We provide a thin adapter (`dise.integrations.hypothesis`) that converts Hypothesis strategies into (D1) distributions, so the same property test can drive both edge-case search and operational reliability estimation.

Summary. DiSE's technical contribution lies between the exact PSE line, which it does not displace where PSE applies, and the sampling-only literature, which it improves upon via symbolic stratification. No prior algorithm combines SMT-driven adaptive partitioning, closed-form mass computation on axis-aligned (D1) regions, and an anytime-valid empirical-Bernstein bound in a single framework.

4 The DiSE Algorithm

We describe DiSE, an adaptive partitioning algorithm for certified reliability estimation. Section 4.1 introduces the frontier data structure. Section 4.2 states the central decomposition theorem, which is the load-bearing correctness result of the paper. Section 4.3 gives the driver loop.

4.1 The Frontier

DiSE maintains a *frontier*: a tree of path-condition regions $\{R_\pi\}_{\pi \in \Pi}$ whose leaves form a partition of $\mathcal{X}$. Each leaf π carries four pieces of data:

- a symbolic region formula F_π in the theory of linear integer/bitvector arithmetic extended with equality and propositional connectives;
- a probability estimate $\hat{w}_\pi \approx \Pr_{\mathcal{D}}[X \in R_\pi]$;
- a sample-based hit-rate estimate $\hat{p}_\pi \approx \Pr[\varphi(P(X)) = 1 \mid X \in R_\pi]$;
- a lifecycle status in $\{\mathsf{OPEN}, \mathsf{CLOSED\text{-}T}, \mathsf{CLOSED\text{-}F}, \mathsf{EMPTY}\}$.

A leaf is OPEN if its hit rate is not yet symbolically certified. It is CLOSED-T or CLOSED-F if the SMT backend has proved that $\varphi(P(x)) = 1$ or $\varphi(P(x)) = 0$ holds uniformly on the region. It is EMPTY if SMT has shown $F_\pi \equiv \bot$. The overall reliability estimator is

$$\hat{\mu} \;=\; \sum_{\pi \in \Pi} \hat{w}_\pi \cdot \hat{p}_\pi. \tag{4}$$

Lemma 1 (Mass-conservative refinement). *When* DiSE *refines a leaf π on a clause b, producing children $\pi \wedge b$ and $\pi \wedge \neg b$, the children's estimated masses satisfy $\hat{w}_{\pi\wedge b} + \hat{w}_{\pi\wedge\neg b} = \hat{w}_\pi$ exactly (modulo importance-sampling noise on non-axis-aligned regions, which is bounded in Section 5).*

Proof (Proof sketch). For axis-aligned LIA children, the closed-form mass of each child is a product of marginal-PMF differences over disjoint coordinate ranges, which sum exactly to the parent's mass. For general regions, DiSE draws one importance-sampling batch from $\mathcal{D} \mid R_\pi$, partitions it by b, and splits $\hat{w}_\pi$ proportionally: $\hat{w}_{\pi\wedge b} = \hat{w}_\pi \cdot \hat{p}_{b|\pi}$ and symmetrically for $\neg b$. The split conserves $\hat{w}_\pi$ by construction.

Lemma 2 (Closure soundness). *Suppose the SMT oracle returns* unsat *for $F_\pi \wedge \neg \mathrm{path}_\pi$, where path_π is the conjunction of branch clauses observed at π. Then every $x \in R_\pi$ traverses path_π in P. If additionally every observation at π agrees on the same φ-value $v \in \{0, 1\}$, then $\varphi(P(x)) = v$ for every $x \in R_\pi$.*

Proof (Proof sketch). Suppose some $x \in R_\pi$ takes a different path. Then some branch clause b_j of path_π is violated by x, so $x \models F_\pi \wedge \neg \mathrm{path}_\pi$, contradicting the oracle's unsat verdict (by soundness of the SMT backend). Determinism of P then forces $\varphi(P(x)) = v$ uniformly on R_π.

4.2 The Central Decomposition Theorem

Let Π_T denote the frontier at the algorithm's stopping time T, decomposed as $\Pi_T = \Pi^{\mathrm{open}} \cup \Pi^{\mathrm{closed}} \cup \Pi_T^{\mathsf{EMPTY}}$. Define the two half-width terms:

$$W_{\mathrm{open}}(T) \;=\; \sum_{\pi \in \Pi^{\mathrm{open}}} \hat{w}_\pi, \tag{5}$$

$$\varepsilon_{\mathrm{stat}}(T) \;=\; \sum_{\pi \in \Pi^{\mathrm{open}}} \hat{w}_\pi \cdot W_\pi(T), \tag{6}$$

where $W_\pi(T)$ is an anytime-valid $(1-\delta_\pi)$-half-width for the per-leaf hit rate, with Bonferroni allocation $\delta_\pi = \delta/K_T$ over the $K_T \le K_{\max}$ leaves created by the algorithm.

Theorem 1 (Central decomposition; coverage at $1-\delta$). *Under Assumption 1, Lemma 1, Lemma 2, soundness of the SMT oracle, and an anytime-valid per-leaf bound W_π,*

$$\mathbb{P}\Big[\,\big|\hat{\mu}_T - \mu\big| \;\le\; \varepsilon_{\text{stat}}(T) + W_{\text{open}}(T)\Big] \;\ge\; 1-\delta. \tag{7}$$

Proof (Proof sketch). Decompose the estimation error:

$$\hat{\mu}_T - \mu = \sum_{\pi\in\Pi^{\text{open}}} \big(\hat{w}_\pi \hat{p}_\pi - w_\pi p_\pi\big) + \sum_{\pi\in\Pi^{\text{closed}}} \big(\hat{w}_\pi - w_\pi\big) p_\pi.$$

The closed-leaf term vanishes for axis-aligned closed leaves (by the closed-form mass) and is bounded by Lemma 1 otherwise. Each open-leaf term splits into a deterministic part—bounded in absolute value by $\hat{w}_\pi$, summing to $W_{\text{open}}(T)$—and a stochastic part bounded with probability $\ge 1-\delta_\pi$ by $\hat{w}_\pi W_\pi(T)$. A union bound over the K_T leaves controls the total stochastic deviation by $\varepsilon_{\text{stat}}(T)$ at confidence $1-\delta$. The anytime property of W_π makes the union bound valid at the data-dependent stopping time T.

Theorem 1 is the load-bearing result. Every implementation choice in Section 5 is justified by reference to it, and every action the algorithm takes reduces either $\varepsilon_{\text{stat}}$ or W_{open}. The reported interval is the two-sided projection

$$[L,U] \;=\; \big[\max(0,\, \hat{\mu}_T - \varepsilon_{\text{stat}}(T) - W_{\text{open}}(T)),\; \min(1,\, \hat{\mu}_T + \varepsilon_{\text{stat}}(T) + W_{\text{open}}(T))\big]. \tag{8}$$

4.3 The Driver Loop

At each iteration, DiSE considers two action families.

Sample(π, b)**.** Draw b additional iid samples from $\mathcal{D} \mid R_\pi$, increment n_π by b, update $\hat{p}_\pi$ and W_π. Expected gain: $\Delta\varepsilon_{\text{stat}} \approx \hat{w}_\pi \cdot (W_\pi(n_\pi) - W_\pi(n_\pi + b))$. Cost: b program evaluations.

Refine(π, b)**.** Select a branch clause b observed at π. Split π into $\pi\wedge b$ and $\pi\wedge\neg b$, distributing $\hat{w}_\pi$ per Lemma 1, then immediately attempt closure on each child (Lemma 2). Expected gain: $\Delta W_{\text{open}} \approx \hat{w}_\pi \cdot \Pr[\text{at least one child closes}]$. Cost: one SMT call (split) plus two (closure checks).

The scheduler selects $\arg\max_a \text{gain}(a)/\text{cost}(a)$, where cost is normalised to one concolic run per unit. The algorithm terminates when $\varepsilon_{\text{stat}}(T) + W_{\text{open}}(T) \le \varepsilon$, or when one of three optional caps fires: sample budget exhausted, wall-clock limit reached, or no action exceeds a minimum gain-per-cost floor.

DiSE is parametrised by three engineering choices: the candidate-clause heuristic for Refine, the per-leaf bound construction W_π, and the SMT backend. Each choice is discussed in Section 5.

Fig. 1. The DiSE adaptive partitioning algorithm

Input: program P, distribution $\mathcal{D}$, property φ, target (ε, δ), optional budget B.
Output: estimator $\hat{\mu}$ and certified interval $[L, U]$.

$\Pi \leftarrow \{\pi_0\}$ with $F_{\pi_0} = \top$, $\hat{w}_{\pi_0} = 1$ // single OPEN leaf at root
bootstrap n_{boot} samples from $\mathcal{D}$ at π_0; call TryClose(π_0)
while not Terminated() **do**
  $\mathcal{A} \leftarrow \{\text{Sample}(\pi, b), \text{Refine}(\pi, b) : \pi \in \Pi^{\text{open}}\}$
  $a^\star \leftarrow \arg\max_{a \in \mathcal{A}} \text{gain}(a)/\text{cost}(a)$
  execute $a^\star$; update Π
  call TryClose(π) for every newly-affected leaf
end while
return $(\hat{\mu}, [L, U])$ per (4) and (8)

5 Implementation

DiSE is an open-source Python prototype of approximately 5,000 lines of code with 290+ tests. This section covers the three engineering choices that turn Section 1 into a working artifact: the concolic front-end, the per-leaf bound W_π, and the SMT abstraction layer.

5.1 Concolic Execution and Branch Capture

A concolic run [?,?] executes $P(x)$ on a concrete input x while simultaneously tracking the symbolic conditions on x that drove each branch taken. DiSE implements this via a tagged-integer wrapper: arithmetic operations on tagged values propagate their symbolic representations, and comparisons emit a branch record (clause, direction) before delegating to the concrete operator.

Each run produces the path condition (the conjunction of branch clauses along the path taken), the property value $\varphi(P(x))$, and the concrete input. Property evaluation is intentionally *outside* the tracer: φ is evaluated on the concrete output, and the resulting Boolean is appended to the run record. This division of labour keeps path conditions small, since φ is typically a thin projection ($y \leq k$, `popcount(y)` ≥ 0) that should not inflate the symbolic representation.

5.2 The Per-Leaf Bound W_π

Theorem 1 requires an anytime-valid $(1 - \delta_\pi)$-half-width for each open leaf's hit rate. DiSE exposes four choices via the `method` parameter.

Wilson (`method="wilson"`). The classical fixed-n Wilson score interval [?]. Tight on Bernoulli leaves at a pre-committed sample count; *not* sound under data-dependent stopping. Appropriate only when the budget is fixed in advance.

Anytime Wilson (`method="anytime"`). The Wilson interval evaluated at confidence $1-\delta_n$ with $\delta_n = 6\delta_\pi/(\pi^2 n^2)$, union-bounded over all n via $\sum_{n=1}^{\infty} 6/(\pi^2 n^2) = 1$. Sound under data-dependent stopping; loose by a $\sqrt{\log n}$ factor and a $\pi^2/6$ inflation.

Predictable-plug-in empirical Bernstein (`method="betting"`). Theorem 2 of Waudby-Smith and Ramdas [?]. For a Bernoulli history $X_1, \ldots, X_n$ at leaf π, the half-width is

$$W_\pi \;=\; \frac{\log(2/\delta_\pi) + \sum_{i=1}^{n} v_i\, \psi_E(\lambda_i)}{\sum_{i=1}^{n} \lambda_i}, \qquad \psi_E(\lambda) = \tfrac{-\log(1-\lambda)-\lambda}{4}, \tag{9}$$

with $v_i = 4(X_i - \hat{\mu}_{i-1})^2$ and predictable bet

$$\lambda_t \;=\; \min\!\left(\sqrt{\frac{2\log(2/\delta_\pi)}{\hat{\sigma}_{t-1}^2 \cdot t\log(t+1)}},\; c\right), \quad c \in (0,1), \tag{10}$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t^2$ are Wilson-regularised running mean and variance. The default truncation is $c = 1/2$, as recommended in [?, §3.2]. PrPl-EB is anytime-valid, closed-form, and variance-adaptive: its asymptotic width is $\sqrt{2\sigma^2 \log(2/\delta_\pi)}$, matching Bennett's inequality without requiring known variance. This is the recommended default for certified results and is strictly tighter than `anytime` on low-variance leaves.

Classical and Maurer–Pontil Bernstein. Provided as `bernstein` and `empirical-bernstein` for baseline comparison only; both are fixed-n bounds and not sound under data-dependent stopping.

5.3 The Closure Rule

A leaf is closed only when the SMT backend certifies path-determinism:

```
def try_close(node):
    if n_samples < min_samples: return False
    if not all observed paths identical: return False
    if not all observed phi values identical: return False
    if is_satisfiable(F_pi AND NOT observed_path) != "unsat":
        return False
    node.status = CLOSED_T if phi == 1 else CLOSED_F
    return True
```

There is no sample-based fallback. A leaf whose closure the SMT backend cannot certify (`sat` or `unknown`) remains open and contributes to W_{open}. This makes the soundness contract a single lemma (Lemma 2), at the cost of wider intervals on programs whose path structure is symbolically intractable.

5.4 SMT Abstraction

The closure check, region-emptiness check, and refinement-clause selection all go through an `SMTBackend` protocol with three implementations: a Z3 backend [?] for production use, a `MockBackend` restricted to axis-aligned LIA that permits the test suite to run without external dependencies, and a `CachedBackend` wrapper that memoises solver queries (typical cache hit rate: 40–60% on the benchmark suite). All backends guarantee soundness of `sat`/`unsat`; returning `unknown` is always admissible.

5.5 Refinement-Clause Selection

DiSE's Refine action requires selecting a branch clause b from the finite set harvested from observed paths at π. DiSE scores each candidate by expected mass balance—intuitively, the clause that partitions $\hat{w}_\pi$ closest to 50/50, maximising the probability that at least one child closes. The heuristic is deliberately conservative; a principled alternative based on Craig interpolation is identified as the primary next development direction (Section 7).

5.6 Reproducibility

DiSE's source code, benchmark suite, and reproduction scripts are available at `https://github.com/<anon>/dise` (URL anonymised for review). The Docker image and `scripts/reproduce.sh` regenerate all results in Section 6 in approximately 30–60 minutes on commodity hardware. Seed control is centralised: each `(method, benchmark, seed)` triple is fully deterministic.

6 Experimental Evaluation

We evaluate DiSE on a suite of 12 integer benchmarks against two baselines: *Plain Monte Carlo* with Wilson-score intervals, and *Stratified Random Monte Carlo* with 16 random hash buckets (Bonferroni-corrected). We report half-width at a fixed sample budget, sample efficiency to a target half-width, and empirical coverage of the certified interval.

6.1 Benchmarks

Table 1 describes the suite. The benchmarks span three regimes: *LIA-uniform*, where PSE would also apply; *LIA-non-uniform*, where path probabilities are non-trivial products of marginal PMF values under the operational distribution; and *non-LIA*, including multiplication, modular exponentiation, and data-structure depth. One benchmark targets the assertion-violation framing.

Table 1. Benchmark suite. $\mu^\star$ is the high-precision Monte Carlo reference used for soundness verification. "Regime" characterises the path-condition arithmetic.

Name	Property	$\mu^\star$	Regime
`coin_machine_U(1,9999)`	output $= 1$	0.0099	LIA, uniform, rare event
`gcd_steps_le_5_BG(0.1,100)`	steps ≤ 5	0.989	LIA, BoundedGeometric
`modpow_fits_in_4b_m=37`	output < 16	0.62	non-LIA, uniform
`miller_rabin_w=2_BG(0.05,200)`	accepts	0.91	non-LIA, BoundedGeometric
`popcount_w6`	≥ 0 (tautology)	1.000	bitvector, uniform
`parity_w6`	$\in \{0, 1\}$	1.000	bitvector, uniform
`log2_w6`	≥ -1	1.000	bitvector, uniform
`collatz_le_30_BG(0.05,200)`	steps ≤ 30	0.74	non-LIA, BoundedGeometric
`sieve_primality_U(2,200)`	is prime	0.23	LIA, uniform
`integer_sqrt_correct_U(1,1023)`	spec holds	1.000	non-LIA, uniform
`sparse_trie_depth_le_3_U(0,63)`	depth ≤ 3	1.000	data-structure, uniform
`assertion_overflow_mul_w=8_U(1,31)`	$a \cdot b$ overflows	0.401	non-LIA, assertion

Table 2. Median certified half-width at shared budget 2,000, 3 seeds, Z3 backend, $\delta = 0.05$, `method="betting"`. **Bold** = tightest. Parentheses show empirical coverage of the reported interval against the MC reference.

Benchmark	Plain MC	Stratified MC	DiSE
`coin_machine_U(1,9999)`	0.0041 (100%)	0.0239 (100%)	**0.0107** (100%)
`gcd_steps_le_5_BG(0.1,100)`	**0.0046** (100%)	0.0256 (100%)	0.500 (100%)
`assertion_overflow_mul_w=8_U(1,31)`	**0.0215** (100%)	0.1247 (100%)	— (see Section 6.5)
Remaining benchmarks: see `results/summary.json` (anonymised supplement).			

6.2 Half-Width at Fixed Budget

Table 2 reports median certified half-width across three seeds at a shared budget of 2,000 concolic runs (DiSE) or 2,000 Monte Carlo draws (baselines).

Three observations stand out. First, DiSE achieves the tightest half-width on the rare-event benchmark (`coin_machine`) and on benchmarks whose closure rule applies cleanly (`popcount_w6`, `parity_w6`): symbolic stratification concentrates the sample budget in regions of high variance while computing closed-form mass elsewhere. Second, plain Monte Carlo dominates on benchmarks whose operational distribution is well-mixed across many paths and whose reliability is near $\mu \approx 1/2$: when DiSE's frontier fragments into many high-mass open leaves, the Bonferroni correction over leaf count washes out the structural-variance gain. Third, stratified random Monte Carlo with 16 buckets is rarely competitive, paying the Bonferroni cost without symbolic guidance that concentrates mass in informative leaves.

Convergence plot placeholder: median half-width vs. sample budget on `coin_machine_U(1,9999)`, 3 seeds, log–log axes; DiSE (`method="betting"`), Plain MC, Stratified MC.

Fig. 2. Convergence on the rare-event benchmark. DiSE reaches half-width ≤ 0.005 in ≈ 200 concolic runs; plain Monte Carlo requires $\geq 10^5$. Stratified random Monte Carlo lies consistently in between.

6.3 Sample Efficiency to a Target Half-Width

A complementary view fixes the target half-width and measures the sample count required to reach it. For `coin_machine_U(1,9999)` at $\varepsilon = 0.005$, $\delta = 0.05$: plain Monte Carlo requires $\Theta(1/(\mu\varepsilon)) \approx 10^5$ samples; DiSE reaches the same target in approximately 200 concolic runs by refining the input space into four path-condition leaves and computing closed-form mass on each.

Figure 2 plots median half-width versus samples used for this benchmark.

6.4 Empirical Coverage

For each `(method, benchmark, seed)` triple we record whether the Monte Carlo reference $\mu^\star$ falls within the reported $[L, U]$. Aggregated over all seeds and benchmarks, DiSE's empirical coverage is at least $1 - \delta$ on every benchmark in the suite *except* `assertion_overflow`; see Section 6.5.

6.5 A Measured Soundness Regression

On `assertion_overflow_mul_w=8_U(1,31)`, DiSE with the strict closure rule of Section 5.3 reports $\hat{\mu} = 0.4300$ with half-width 0—a certified point estimate approximately 0.03 away from the true value 0.4006. Empirical coverage is 0% across three seeds. The failure is independent of the per-leaf bound choice and the SMT backend.

The root cause is a gap in Lemma 2: the lemma proves that every $x \in R_\pi$ traverses the same program path, which is sufficient when φ is determined by the path alone, but not when φ depends on the specific output value along that path. For the benchmark `a*b < 256` with $\varphi(y) = [y \geq 256]$, the property comparison occurs outside the concolic tracer; closing a leaf on symbolic path-determinism leaves the per-input property value uncertified. The fix—either routing property comparisons through the tracer, or augmenting the closure rule to certify φ-constancy directly—is straightforward to state and implement. We defer it to follow-up work and report the benchmark as a known limitation.

6.6 Discussion

The empirical picture is consistent with the theoretical analysis of Section 3. DISE performs best when the operational distribution concentrates on path-deterministic regions of high probability mass: rare-event programs, bitvector kernels where all inputs take the same path, and benchmarks with clean LIA path structure under non-uniform distributions. It ties plain Monte Carlo on high-variance, well-mixed benchmarks where symbolic stratification offers little variance reduction, and is dominated by plain Monte Carlo when the SMT cost of refinement is not offset by the closure rule. These regimes align directly with the upper-bound argument sketched in Section 4.

7 Conclusion

We have presented DISE, an algorithm for distribution-aware reliability estimation that maintains a frontier of path-condition regions and at each step chooses between sampling and SMT-driven refinement under a unified gain-per-cost criterion. The certified half-width decomposes into a statistical component $\varepsilon_{\text{stat}}$, driven down by sampling, and a structural component W_{open}, driven down by symbolic refinement. This decomposition—Theorem 1—is simultaneously the algorithm's correctness argument and its scheduler. Two implementation choices distinguish DISE from prior work: the per-leaf statistical bound instantiates the closed-form predictable-plug-in empirical-Bernstein confidence sequence of Waudby-Smith and Ramdas, and the closure rule is single-branch (SMT-only), yielding a one-lemma soundness contract and eliminating a class of silent bias that affects prior prototypes.

Limitations. We identify five.

1. *Path-determinism versus property-constancy.* Section 6.5 reports a measured soundness regression on a non-LIA benchmark where the property is evaluated outside the concolic tracer. The fix is well-understood but not yet implemented.
2. *Distribution class.* The (D1) restriction excludes Bayes-net-structured and general discrete distributions. Lifting it requires ancestor-conditioned constrained samplers and a mass estimator that handles coordinate dependence.
3. *Floating-point arithmetic and symbolic memory.* Both are out of scope for the current prototype.
4. *Refinement-clause selection.* The current heuristic picks from clauses observed in samples; it does not synthesise new predicates. Craig interpolation is the natural principled alternative.
5. *Statistical efficiency under partition fragmentation.* When the frontier fragments into many high-mass open leaves, the Bonferroni correction inflates $\varepsilon_{\text{stat}}$ in proportion to the leaf count, and the SMT cost of refinement is not recovered by variance reduction.

Future work. Two extensions deserve immediate attention. First, the closure rule's implicit path-implies-property assumption can be discharged by routing φ through the concolic tracer—straightforward, requiring only careful handling of short-circuit evaluation. Second, replacing the candidate-clause scorer with Craig interpolation [?,?] would let the algorithm synthesise predicates the program never explicitly branches on, broadening the class of programs for which W_{open} can be driven to zero. Beyond these, we are interested in substituting a certified approximate model counter [?,?] for the importance-sampling estimator on non-axis-aligned regions—yielding machine-checkable certificates end-to-end—and in extending the distribution class to handle (D2) Bayes-net-structured inputs.

Reproducibility. DISE, including all benchmarks, scripts, and the Docker image, is anonymised for review at `https://github.com/<anon>/dise`. All tables and figures in this paper are reproduced end-to-end by `scripts/reproduce.sh`.